(Please write your Exam Roll No.)

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: ETCS-206 **Subject: Theory of Computation**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Assume missing data, if any.

Q1 Differentiate between the following: **[5x5=25]**

a) Regular Language and Context Free language
b) Recursive and Recursive Enumerable language
c) Halting Problem & Un-decidable problem of Turing machine
d) Classify & Differentiate between different types of FA with output.
e) Dead state and In-accessible state

Q2 a) Explain closure properties of regular sets. Provide at least two closure properties of regular sets and justify their significance in formal language theory. **[4]**

b) Convert the following regular expression to DFA.
ab(a+b)*ab **[4]**

c) Explain the significance of the stack in a Pushdown Automaton with example. **[4.5]**

Q3 a) How the ambiguity of a grammar can be removed? Explain with suitable example. **[6]**

b) How to Convert any CFG to GNF? Explain with suitable example. **[6.5]**

Q4 a) What are Normal Forms in grammar? Explain its importance with example. **[4]**

b) Explain two types of acceptance in pushdown automata for a^nb^n. **[4.5]**

c) Define Pushdown Automata and explain its components. **[4]**

Q5 Write a short note on following

a) Kleen's closure & Positive Closure **[4]**
b) Undecidability **[4.5]**
c) SPACE and PSPACE **[4]**

Q6 a) Write the steps and convert the regular grammar into Finite Automata for **[6.5]**
$S \rightarrow 0A/1B0/1$ $A \rightarrow 0S/1B/1$ $B \rightarrow 0A/S$

b) Construct a TM for a^nb^n. **[6]**

Q7 a) Design a Turing machine for string of palindrome (Same machine for even and odd length). **[7.5]**

b) Write a short note on Computability and Intractability. **[5]**

P.T.O.

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